Optimally Measured Intrinsic Parameters of Gravitational-Wave Sources: Generalisations of $\chi_{\rm eff}$ and Chirp Mass

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Abstract

We determine the linear combinations of a compact-binary's intrinsic parameters that are *best measured* by a gravitational-wave detector network, as smooth functions of the true source parameters. Working in the Fisher information approximation, we marginalise over all other parameters, whiten the resulting covariance so that parameters of different dimension are comparable, and diagonalise. The eigenvector of smallest variance defines the best-measured combination. For aligned spins this combination reproduces the effective spin $\chi_{\rm eff}$ to leading order and we provide a closed-form generalisation. For the masses, expressed in $(\log m_1, \log m_2)$, the best-measured combination smoothly interpolates between the chirp mass at low total mass and the total mass at high total mass.

1 Setup and conventions

We consider a quasi-circular, aligned-spin compact binary. The frequency-domain waveform $h(f;\boldsymbol{\theta})$ depends on the parameter vector

$$\boldsymbol{\theta} = \big(\mathcal{M},\, \eta,\, d_L,\, \theta,\, \phi,\, \iota,\, \psi,\, t_c,\, \phi_c,\, \chi_{1z},\, \chi_{2z}\big), \tag{1}$$

with detector-frame chirp mass $\mathcal{M}$, symmetric mass ratio $\eta = m_1 m_2/(m_1+m_2)^2$, luminosity distance d_L, sky position (θ,ϕ), inclination ι, polarisation ψ, coalescence time and phase (t_c,ϕ_c), and the dimensionless aligned spin components χ_{1z},χ_{2z}. Source-frame masses are redshifted to the detector frame, $m_i^{\rm det} = (1+z)\, m_i^{\rm src}$, and the redshift fixes d_L through a fiducial cosmology. The masses can equivalently be parametrised by (m_1,m_2) or $(\log m_1, \log m_2)$; we use the last of these for the mass study (Sec. 4).

Fisher matrix. For a network of detectors with one-sided noise PSDs $S_n^{(d)}(f)$, the Fisher information matrix is

$$F_{ij} = \sum_d \left(\frac{\partial h}{\partial \theta_i} \,\middle|\, \frac{\partial h}{\partial \theta_j} \right)_d, \qquad (a\,|\,b)_d = 4\,\mathrm{Re}\int_{f_{\rm min}}^{f_{\rm max}} \frac{a^*(f)\, b(f)}{S_n^{(d)}(f)}\, df. \tag{2}$$

The waveform derivatives are evaluated by automatic differentiation (`gwfast`/JAX), giving smooth, low-noise gradients — essential for the eigenvectors to vary smoothly with $\boldsymbol{\theta}$. In the large-SNR (Gaussian-linear) limit the parameter covariance is $\boldsymbol{C} = \boldsymbol{F}^{-1}$.

2 Best-measured directions

Let $\boldsymbol{\theta}_S \subset \boldsymbol{\theta}$ be an intrinsic sub-space of interest (the spins, or the masses). Marginalising over all other parameters under the Gaussian approximation amounts to taking the corresponding sub-block of the full covariance,

$$\boldsymbol{C}_S = \left[\boldsymbol{F}^{-1}\right]_S. \tag{3}$$

A linear combination $p = \boldsymbol{a}^\top \boldsymbol{\theta}_S$ has variance $\sigma_p^2 = \boldsymbol{a}^\top \boldsymbol{C}_S\, \boldsymbol{a}$. The best-measured combination minimises this variance, so it is the eigenvector of $\boldsymbol{C}_S$ with the *smallest* eigenvalue.

Whitening. Because the parameters carry different units and natural scales (e.g. $\mathcal{M}$ vs. η), the raw eigenproblem is scale-dependent. We whiten by the marginal standard deviations $s_i = \sqrt{(\boldsymbol{C}_S)_{ii}}$, i.e. work in the dimensionless variables $\boldsymbol{u} = \boldsymbol{D}^{-1}\boldsymbol{\theta}_S$ with $\boldsymbol{D} = \mathrm{diag}(s_i)$. The whitened covariance is the correlation matrix

$$\boldsymbol{C}_S^w = \boldsymbol{D}^{-1}\boldsymbol{C}_S\, \boldsymbol{D}^{-1}, \tag{4}$$

which we symmetrise and diagonalise, $\boldsymbol{C}_S^w\, \boldsymbol{b}_j = \lambda_j\, \boldsymbol{b}_j$, with $\lambda_1 \le \lambda_2 \le \ldots$ ordered ascending. The whitened eigenvectors $\boldsymbol{b}_j$ are orthonormal. The corresponding combination in the original parameters has coefficients

$$\boldsymbol{a}_j = \boldsymbol{D}^{-1}\boldsymbol{b}_j \;=\; \boldsymbol{b}_j/\boldsymbol{s}, \tag{5}$$

which we normalise to unit Euclidean norm. The leading vector $\boldsymbol{a}_1$ is the best-measured combination; $\boldsymbol{a}_2$ is the next-best, and so on.

Numerical reductions. Three equivalent-in-principle reductions are implemented to assess numerical stability: (i) *marginalise* — invert the full Fisher, take the sub-block [Eq. (3)]; (ii) *whiten-then-marginalise* — whiten the full Fisher by its diagonal before inversion; (iii) *condition* — fix all other parameters and invert the Fisher sub-block. We adopt (i) by default.

Smoothness across the grid. The sign and ordering of eigenvectors are arbitrary at each point. We track each mode along the parameter grid by matching the orthonormal whitened eigenvectors $\boldsymbol{b}_j$ between adjacent points (maximising overlap and fixing the sign for positive overlap), which is robust through eigenvalue crossings. Matching the non-orthonormal physical coefficients $\boldsymbol{a}_j$ would mis-assign near crossings.

3 Spin: generalisation of $\chi_{\rm eff}$

The effective aligned spin is

$$\chi_{\rm eff} = \frac{\chi_{1z} + q\,\chi_{2z}}{1+q}, \qquad q \equiv \frac{m_2}{m_1} \le 1, \tag{6}$$

i.e. the combination of (χ_{1z}, χ_{2z}) with weight ratio $c_2/c_1 = q$. We compute the best-measured spin direction in the (χ_{1z}, χ_{2z}) sub-space by the procedure of Sec. 2 and report its weight ratio c_2/c_1.

Numerically the best-measured direction tracks $\chi_{\rm eff}$ closely but lies systematically slightly below it. A two-parameter fit (single LIGO at design sensitivity, `IMRPhenomD`; weak dependence on the other parameters) gives

$$\boxed{\left(\frac{c_2}{c_1}\right)_{\rm best} = q\Big[\,1 - 0.337\,(1-q) + 0.359\,(1-q)^2\,\Big],} \tag{7}$$

with RMS residual ≈ 0.018 and a maximum deviation from the $\chi_{\rm eff}$ prediction $c_2/c_1 = q$ of ≈ 0.10 near $q \sim 0.5$. Equation (7) reduces to $\chi_{\rm eff}$ at equal mass ($q \to 1$) and stays within $\sim 10\%$ of it across the range, quantifying the (modest) sense in which $\chi_{\rm eff}$ is the optimally measured aligned-spin parameter.

4 Mass: chirp mass to total mass

We study the masses in the basis $(\log m_1, \log m_2)$, in which the best-measured combination $w_1 \log m_1 + w_2 \log m_2$ is a power-law mass parameter $\propto m_1^{w_1} m_2^{w_2}$ and is scale-free. The Fisher matrix in this basis is obtained from the (m_1, m_2) Fisher by the diagonal Jacobian $\boldsymbol{F}_{\log} = \boldsymbol{JFJ}$ with $\boldsymbol{J} = \mathrm{diag}(m_1, m_2, 1, \dots)$.

Two reference combinations bracket the expected behaviour. The chirp mass

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}, \qquad \log \mathcal{M} = \tfrac{3}{5}\big(\log m_1 + \log m_2\big) - \tfrac{1}{5}\log(m_1 + m_2), \tag{8}$$

has gradient weights in $(\log m_1, \log m_2)$ with ratio

$$\left(\frac{w_2}{w_1}\right)_{\mathcal{M}} = \frac{\partial \log \mathcal{M}/\partial \log m_2}{\partial \log \mathcal{M}/\partial \log m_1} = \frac{3 + 2q}{2 + 3q}, \tag{9}$$

while the total mass $M_{\rm tot} = m_1 + m_2$ has

$$\left(\frac{w_2}{w_1}\right)_{M_{\rm tot}} = \frac{m_2/M_{\rm tot}}{m_1/M_{\rm tot}} = q. \tag{10}$$

At low total mass, where many inspiral cycles lie in band, the phase is dominated by $\mathcal{M}$ and the best-measured combination matches Eq. (9). At high total mass, where the signal is dominated by the merger–ringdown ($f \sim 1/M_{\rm tot}$), it approaches Eq. (10). We model the crossover as

$$\boxed{\left(\frac{w_2}{w_1}\right)_{\rm best} = \big[1 - s(M_{\rm tot})\big]\,\frac{3 + 2q}{2 + 3q} + s(M_{\rm tot})\,q, \qquad s(M_{\rm tot}) = \frac{1}{1 + \big(M_{\rm tot}/M_{\rm tot}^{\star}\big)^{-k}},} \tag{11}$$

a sigmoid in $\log_{10} M_{\rm tot}$ with transition mass $M_{\rm tot}^{\star}$ and sharpness k. The single-detector design fit gives $M_{\rm tot}^{\star} \sim 1.5$–$2.5 \times 10^2\, M_\odot$ for the transition.

Caveats. Equation (11) captures the chirp-to-total trend but is not yet a high-fidelity fit. The data show (i) a non-monotonic *overshoot* near $M_{\rm tot} \sim 100$–$130\, M_\odot$, where w_2/w_1 rises above the chirp value before falling toward the total-mass value — a real feature the simple sigmoid cannot represent; and (ii) ill-conditioning at equal mass ($q \to 1$), where the two $(\log m_1, \log m_2)$ eigenvalues become near-degenerate and the best-measured mode can spuriously flip to the antisymmetric direction $w_2/w_1 \to -1$ (the poorly measured mass-ratio direction). A richer functional form and explicit regularisation of the $q = 1$ degeneracy are required for a quantitative mass formula; the qualitative chirp→total transition is robust.

5 Summary

The best-measured intrinsic parameters, defined as the smallest-variance eigenvectors of the marginalised, whitened Fisher covariance, generalise the familiar phenomenological combinations:

- **Spin:** a smooth, $\sim 10\%$ correction to $\chi_{\rm eff}$, Eq. (7), reducing to $\chi_{\rm eff}$ at equal mass.
- **Mass:** a power-law mass parameter that interpolates between chirp mass [Eq. (9)] and total mass [Eq. (10)] across a transition near $M^{\star}_{\rm tot} \sim 10^2\,M_{\odot}$, Eq. (11).

These are derived for aligned spins with the `IMRPhenomD` waveform and a single LIGO design-sensitivity detector; the framework is configurable for arbitrary detector networks. Precessing spins (and the corresponding generalisation of χ_p) require a precessing waveform model and are left to future work.